Quality Control

QC Data Management Solutions

Introduce your laboratory to a whole new world of Unity data management solutions

Visit **www.bio-rad.com/qualitycontrol** for more information.

BIO-RAD

Commitment to Quality Control

Using our comprehensive product line of independent controls is an important step toward helping you improve the reliability of your laboratory results. The ability to successfully manage and interpret your quality control results is also essential to producing solid, dependable laboratory outcomes.

Bio-Rad Laboratories offers a vision and a commitment to quality control that is unsurpassed in the clinical laboratory industry. Today, your laboratory can be connected to the Unity Interlaboratory Program, the world's largest community of quality control users. Improve patient test results and cost effectiveness with quality control products, quality control data management solutions, industry-leading interlaboratory comparisons and personalized service from Bio-Rad.

Regulatory Requirements

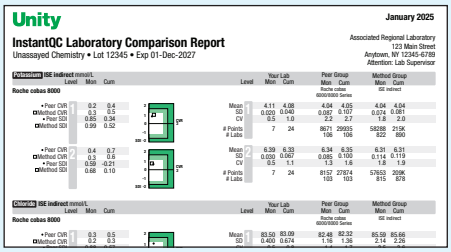
"The laboratory shall monitor its performance of examination methods, by comparison with results of other laboratories."

–ISO 15189:2022(E), Subclause 7.3.7.3

Unity Interlaboratory Program

Participation in an interlaboratory program is extremely important for laboratories to help ensure the reliability and precision of test systems. No program in the world is more powerful and effective at providing information and helping to improve laboratory analytical performance than the Unity Interlaboratory Program.

- Benefit from peer QC data generated from more than 59,000 instruments in 100+ countries
- Recognize analytical process improvement opportunities
- Increase confidence in proficiency survey outcomes
- Identify unrecognized trends or shifts that may occur between proficiency surveys
- Receive on-demand interlaboratory comparison reports
- Receive monthly reports with regular data submissions
- Submit data in real time using one of our Unity software solutions
- Detect and identify potentially important analytical errors
- Meet accreditation and regulatory requirements



Unity

January 2025

InstantQC Laboratory Comparison Report

Unassayed Chemistry • Lot 12345 • Exp 01-Dec-2027

Associated Regional Laboratory
123 Main Street
Anytown, NY 12345-6789
Attention: Lab Supervisor

Potassium ISE indirect mmol/L

	Level	Mon	Cum
Roche cobas 8000			
• Peer CVR	1	0.3	0.4
Method CVR		0.3	0.5
• Peer SDI		0.85	0.34
Method SDI		0.99	0.32
• Peer CVR	2	0.4	0.7
Method CVR		0.3	0.6
• Peer SDI		0.59	-0.21
Method SDI		0.68	0.10

	Level	Your Lab Mon	Your Lab Cum	Peer Group Mon (Roche cobas 6000/8000 Series)	Peer Group Cum	Method Group Mon (ISE indirect)	Method Group Cum
Mean	1	4.11	4.08	4.04	4.05	4.04	4.04
SD		0.020	0.040	0.037	0.107	0.074	0.081
CV		0.5	1.0	2.2	2.7	1.8	2.0
# Points		7	24	8671	29955	58288	215K
# Labs				106	106	822	850
Mean	2	6.39	6.33	6.34	6.35	6.31	6.31
SD		0.030	0.067	0.085	0.100	0.114	0.119
CV		0.5	1.1	1.3	1.6	1.8	1.9
# Points		7	24	8157	27874	57653	209K
# Labs				103	103	815	878

Chloride ISE indirect mmol/L

	Level	Mon	Cum
Roche cobas 8000			
• Peer CVR	1	0.3	0.5
Method CVR		0.2	0.3

	Level	Your Lab Mon	Your Lab Cum	Peer Group Mon (Roche cobas 6000/8000 Series)	Peer Group Cum	Method Group Mon (ISE indirect)	Method Group Cum
Mean	1	83.50	83.09	82.48	82.32	85.59	85.66
SD		0.400	0.674	1.18	1.36	2.14	2.26

Receive On-demand InstantQC Reports

Get started at QCNet.com today!

InstantQC Reports allow you to compare your control results to other laboratories at any time. These on-demand reports are particularly useful for troubleshooting problems with test system performance.

The Advantages of Participating in the Unity Interlaboratory Program

Detecting Errors in Your Test System

Without a high quality interlaboratory program, your laboratory may not be aware of gradual or sudden changes in your test system that may be caused by events such as reagent or calibrator reformulations, standardization changes, or instrument software changes. An interlaboratory program may offer early awareness of shifts and trends to help avoid costly test repeats and unnecessary troubleshooting.

Your Proficiency Program is Not Enough

Proficiency surveys provide a snapshot in time and should not be used as a substitute for daily quality control with an interlaboratory program.

- Acceptable performance on the day of the proficiency testing does not guarantee testing reliability every day
- Participation in an interlaboratory program is the recommended way to improve confidence in test system performance every day, especially with an interlaboratory program that offers an option to receive instant interlaboratory feedback on-demand

Some advantages for participating in an interlaboratory QC program are:

- Verifying that a laboratory is producing QC results that are consistent with other laboratories using the same measurement procedure, and thus demonstrating that the laboratory is using the measurement procedure correctly

Detecting and identifying bias in a measurement procedure

- Bias can be caused by events such as reagent or calibrator lot changes or reformulations, changes in calibration traceability to reference systems, or instrument software changes. Comparison of an individual laboratory's QC result to a peer group mean value can identify a trend or shift, or ascertain if other laboratories are experiencing the same changes."

Supplementing PT/EQA programs

- PT/EQA programs verify performance at a point in time. Acceptable performance on the day of PT/EQA testing does not guarantee testing reliability every day because errors in a measurement procedure can occur at any time. In addition, the interlaboratory QC data can be used to investigate a failed or questionable PT/EQA result.
  - CLSI, C24, 4th ed. Statistical Quality Control for Quantitative Measurement Procedures: Principles and Definitio 7.2 Using Interlaboratory Quality Control to Assess a Quality Control Program.

Unity Solutions

Bio-Rad Laboratories offers several options for participation in the Unity Interlaboratory Program. Refer to the product comparison grid below to help determine which solution is the most appropriate for your laboratory.

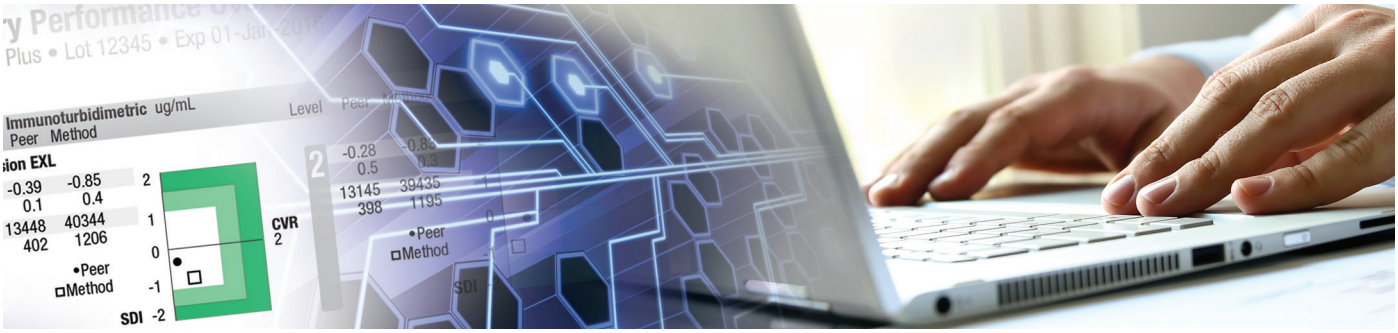
Internet-based solutions such as UnityWeb, Unity Next Peer QC, and Unity Real Time online eliminate the need to install and update software locally in your laboratory, and reduce the amount of support necessary from on-site IT staff.

Desktop software solutions, like Unity Real Time, are appropriate if your internet connection is not sufficient, or if you prefer maintaining local software.

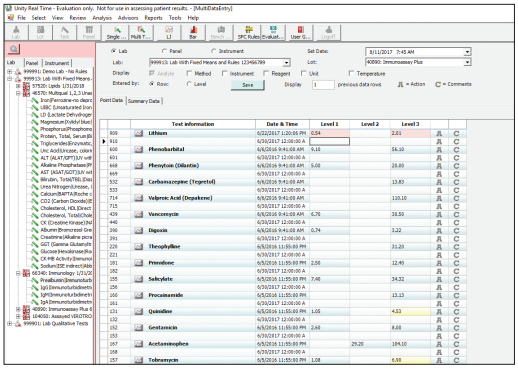
	Basic User Options			Advanced User Options	
	Unity Next Peer QC	UnityWeb	Unity Real Time LT	Unity Real Time Online	Unity Real Time
Platform Design					
Web Service	X	X		X	
Desktop Software			X		X
Bio-Rad Hosted Database	X	X		X	
Single Platform for Data Entry & Peer Reports	X				
Digital Sign & Save for Peer Reports	X				
Unity Interlaboratory Reports (Peer Reports)					
Monthly Evaluation	On-demand	Monthly or Instant QC	Monthly or Instant QC	Monthly or Instant QC	Monthly or Instant QC
Lab Comparison	On-demand	Monthly or Instant QC	Monthly or Instant QC	Monthly or Instant QC	Monthly or Instant QC
Lab Histogram	On-demand	Monthly	Monthly	Monthly	Monthly
Bias and Imprecision		Monthly	Monthly	Monthly	Monthly
Statistical Profile		Monthly	Monthly	Monthly	Monthly
Qualitative Report		Monthly	Monthly	Monthly	Monthly
Affiliated Report	Q3-2025	Monthly	Monthly	Monthly	Monthly
Worldwide report	On-demand	Monthly	Monthly	Monthly	Monthly
Manufacturer Report	On-demand	Monthly	Monthly	Monthly	Monthly
QC Design and Charts					
Westgard Rules	X	X	X	X	X
Levey-Jennigs Chart	X	X	X	X	X
Multi-Levey-Jennings Chart		X		X	X
Additional Features					
Westgard Advisor (Subscription sold separately)		X		X	X
Bench Review		X	X	X	X
Supervisor Review				X	X
Analytical Goals				X	X
Dynamic Data Set Comparisons				X	X
Connectivity (Sold separately)					
WebConnect		X		X	
UnityConnect		X	X	X	X
Unity Next File Upload	X				

Basic Advanced Desktop based Web based Mobile

Want to know more? Discover the power of Unity Solutions at www.bio-rad.com/qc-datamanagement



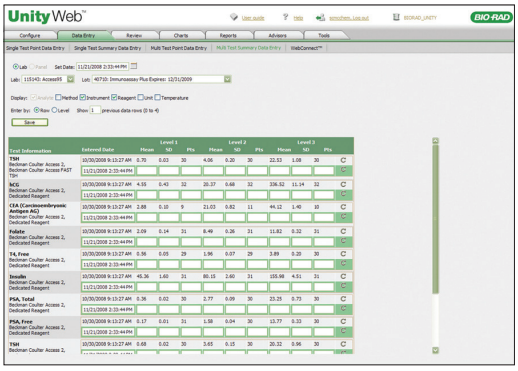
Basic User Options



Unity Real Time LT

Entry-Level Desktop QC Data Management Solution

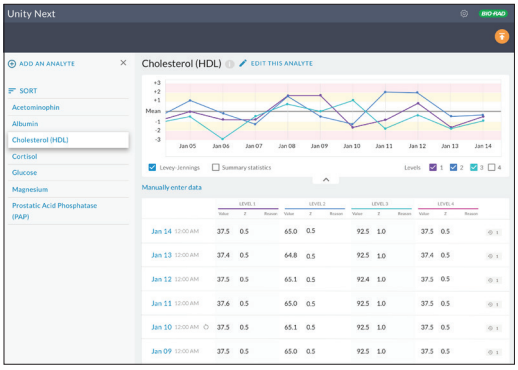
- Basic QC rules, charts and reports
- Configurable Levey-Jennings charts
- Upgrade easily to Unity Real Time for more advanced tools and features
- Upload QC data points from an LIS, middleware or instrument (optional)



UnityWeb

Entry-Level Online QC Data Management Solution

- No software to install and maintain
- Use basic QC rules, charts and reports
- Upgrade easily to Unity Real Time online for more advanced tools and features
- Upload QC data points from an LIS, middleware or instrument (optional)



Unity Next Peer QC

Entry-Level Online QC Data Management Solution with Instant Access to Peer Comparison

- Latest generation of Unity with user-friendly interface
- No software to install and maintain
- Use basic QC rules, Levey-Jennings charts and reports
- Instantly access peer reports for peer comparison and QC troubleshooting
- Upload QC data points from LIS, middleware, or instrument
- Submit QC data and download reports in one location

Connectivity

Basic

WebConnect

File uploads to UnityWeb & Unity Real Time Online Software
A web-based connectivity solution that allows laboratories to easily upload QC data from LIS systems, middleware and/or instruments directly into Unity Real Time online or UnityWeb

- Eliminate manual keying of data
- Standard LIS QC reports can be used
- No software to install

Advanced

UnityConnect

A connectivity solution that allows QC data from LIS systems, middleware and/or instruments to be quickly and easily imported into Unity software, enabling interlaboratory program comparison

- Eliminate manual keying of QC data
- Standard LIS QC reports can be used
- Automatic data import and real-time connection
- Capture and analyze QC data economically from laboratory instruments not connected to the LIS
- Easy-to-install optional interface hardware
- Choose between serial and ethernet connection

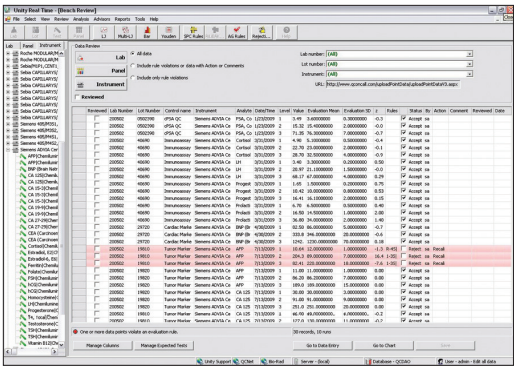
Basic

Unity Next File Upload

File uploads to Unity Next Software

- A data upload solution allowing file uploads of QC data, specifically for Unity Next software
- Upload QC data from LIS, middleware, and/or instruments to Unity Next
- No software to install

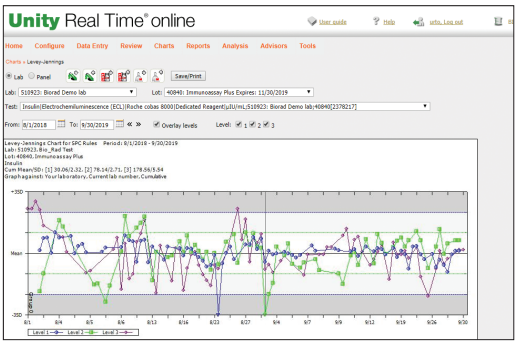
Advanced User Options



Unity Real Time

Expert QC Data Management Solution for Desktop Users

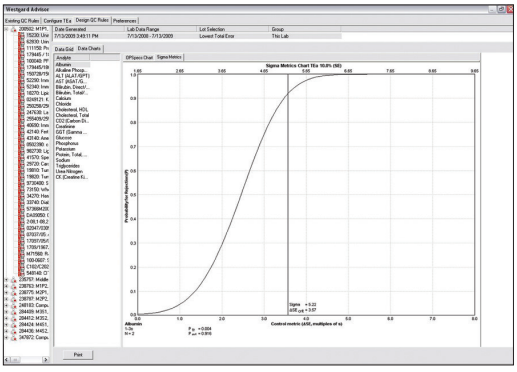
- Facilitate regulatory compliance under CLIA and ISO 15189
- Improve real-time bench and supervisor QC data review
- Implement best QC rules when used with Westgard Advisor
- Run validation with comprehensive audit trails
- Advanced charts and reports for data analysis
- Reduce non-essential retests with Analytical Goal options
- Upload QC data points from an LIS, middleware or instrument (optional)
- RiLiBÄK Advisor module available to comply with German regulations



Unity Real Time online

Expert QC Data Management Solution for Online Users

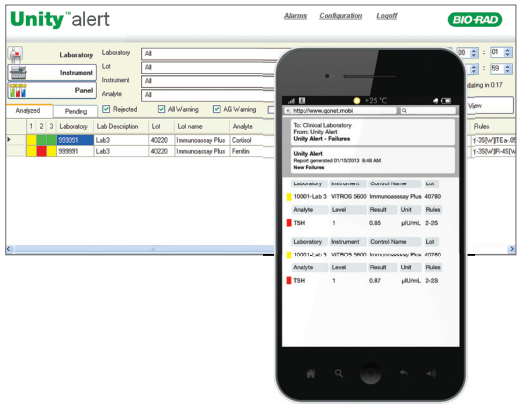
- Facilitate regulatory compliance under CLIA and ISO 15189
- Improve real-time bench and supervisor QC data review
- Implement best QC rules when used with Westgard Advisor
- Run validation with comprehensive audit trails
- Advanced charts and reports for data analysis
- Reduce non-essential retests with Analytical Goal options
- Upload QC data points from an LIS, middleware or instrument (optional)



Westgard Advisor

Automatic QC Rules Selection Engine

- Recommend and automatically apply best QC rules with patented technology
- Easy step-by-step automatic rule selection capabilities
- Reduce false rejections and desensitization to false error flags
- Save time and money by reducing unnecessary repeats and troubleshooting
- Improve laboratory test quality with optimally selected QC rules
- Available as an optional module with Unity QC data management solutions



Unity Alert

Continuously Monitor QC Status

- Notifications of missing QC runs can help to ensure proper monitoring of instrument performance
- Notifications of new violations can be useful for troubleshooting and monitoring problem analytes
- Alert rules are easy to set up and customize for specific instruments or tests
- Receive emails remotely to stay informed of QC issues in your laboratory
- Color-coded displays help draw attention to the most urgent QC issues
- Notifications are provided even when Unity Real Time is not in use
- Available as an optional module with Unity Real Time

Bio-Rad Mission: Control

Take the guesswork out of your QC strategy

Employing a robust quality control system is an essential step for any laboratory hoping to provide reliable results for appropriate patient care. Unfortunately for clinical laboratories, selecting the right QC rules and frequency can be challenging, and questions often arise.

- How often should controls be run for specific assays– is there an optimal frequency?
- How can QC be managed to minimize the risk of releasing incorrect patient results?

Bio-Rad Mission: Control is the first objective risk management solution for quality control in laboratories. The intuitive interface allows users to focus on patient safety by assessing the risk of reporting incorrect patient results in their laboratory.

Bio-Rad Mission: Control allows you to:

- Determine your laboratory's Risk Management Index (RMI)
- Evaluate your test method performance based on your QC results
- Quantify the likelihood and severity of reporting incorrect patient results
- Assess how much you can lower your risk, based on instrument performance
- Create a QC Plan by confidently selecting rules and frequency that fit your risk tolerance

Minimize your risk of reporting unreliable patient results

A patient risk approach to Quality Control is a progressive step toward improving quality assurance and patient care. Bio-Rad can help you understand what this means for your laboratory and how to take steps to implement an effective approach for Quality Control within your analytical test phase.

For more information, visit www.qcnet.com/missioncontrol

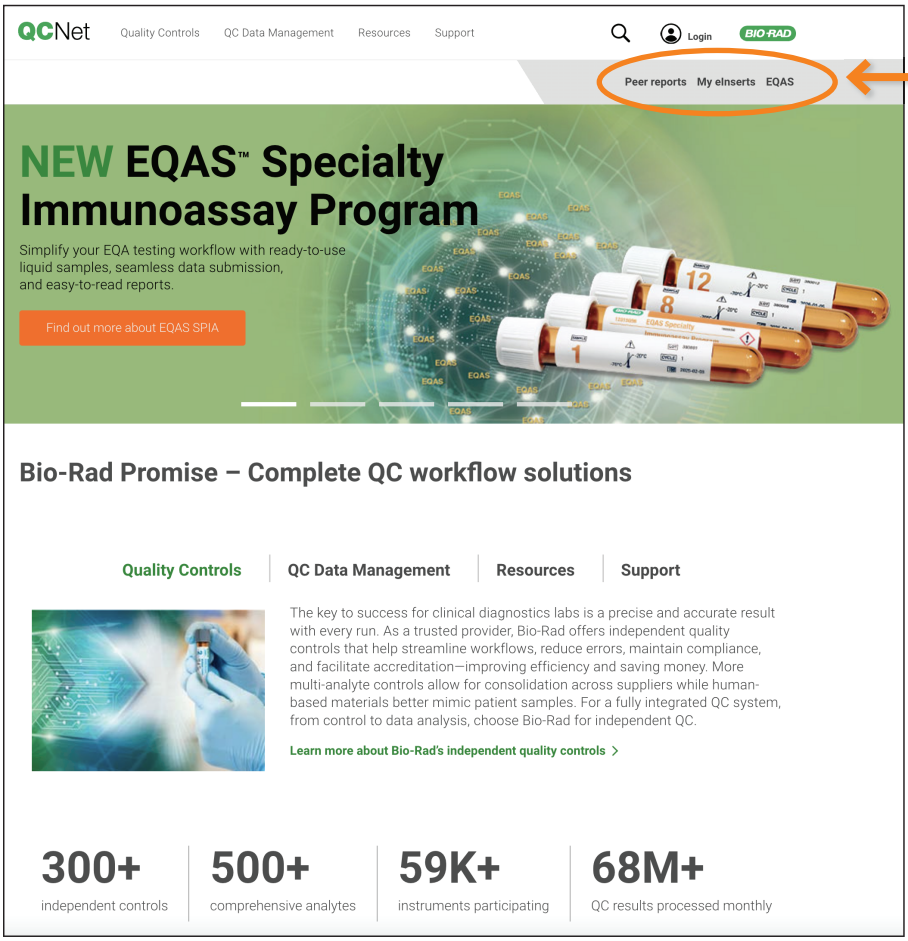
Regulatory Requirements

"IQC shall be performed at a frequency that is based on the stability and robustness of the examination method and the risk of harm to the patient from an erroneous result."

– ISO 15189:2022(E), Subclause 7.3.7.2

QCNet is the Leading QC Portal on the Internet

Learn more about the many QC tools and resources from Bio-Rad at www.QCNet.com



Quick Links to the most-used features

Unity Reports

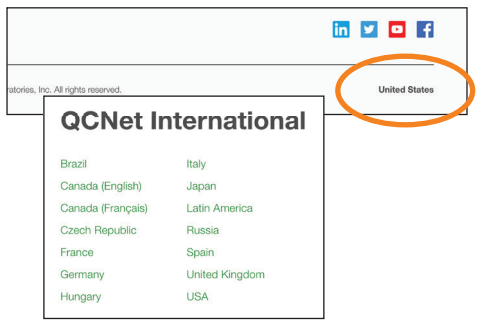
- Get your Unity Reports in just a few clicks.

My eInserts

- Access product inserts

EQAS

- View Reports
- Access EQAS Online
- Training/Tutorials
- View Product Information



Access country-specific information

At the bottom of the QCNet home page, click on the country name shown to access the pop-up menu, then select the desired country*.

*Not all countries are available.

Explore the drop-down menus to find more online QC tools on QCNet

Quality Controls

COVID-91
InteliQ Load and Go
Cardiac Marker
Infectious Disease
Immunoassay
Chemistry
Diabetes & Hemoglobin
Immunology
Blood Gas
Toxicology
Tumor Markers
Therepeutic Drug Monitoring
Hematology
Coagulation
Urinalysis

QC Data Management

UnityWeb
Unity Real Time online
Unity Software Solutions
Peer Reporting Program
Mission: Control
EQAS
Linearity Web
Data Submission Calendar

Resources

Training & Education
Technical Documents
Frequently Asked Questions
Barcode Label Generator
Value Assignment Program

Support

My Account - Log in
Lab Admin
Contact us

Ordering Information

Cat #	Description	
Unity Real Time LT		
805-1	Unity Real Time LT	Annual subscription
825i	Unity Remote Installation & Training	Single use
UnityWeb		
870-1	UnityWeb	Annual subscription
WebConnect		
870-W1	WebConnect 2.0	Annual subscription
825i	Unity Remote Installation & Training	Single use
UnityConnect		
11000009	UnityConnect Software	Annual subscription
858i	Additional Software Training	Single use
Unity Real Time		
804-1	Unity Real Time	Annual subscription
825i	Unity Remote Installation & Training	Single use
Unity Real Time online		
804-W1	Unity Real Time online	Annual subscription

Cat #	Description	
Westgard Advisor		
811-1	Westgard Advisor	Annual subscription
Unity Alert		
806-1	Unity Alert	Annual subscription
825i	Unity Remote Installation & Training	Single use
Bio-Rad Mission: Control		
1100002	Mission: Control	Annual Subscription
1100005	Mission: Control	Subscription and Service Bundle
Unity Next Peer QC		
11000517	Unity Next Peer QC (5 users)	Annual Subscription
11000506	Unity Next Onboarding	Single Use
Unity Next File Upload		
11001897	Unity Next File Upload	Annual Subscription

EQAS
An independent, external assessment of laboratory performance in comparison to your peers.
QCNet.com/eqas

Independent QC
Ongoing, proactive, unbiased daily QC that helps identify errors as they occur or begin to trend.
QCNet.com/independentqc

Unity
QC Data Management tools that help you create a strategy to reduce risk and streamline QC workflow.
QCNet.com/datamanagement

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Bio-Rad Laboratories, Inc.

Clinical Diagnostics Group

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